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pendulum. This is similar to free fall. Therefore, in a pendulum of fixed length, it does not matter whether an elephant swings or an ant swings. Both of them will swing with the same time period.

(ii) Amplitude of the oscillations

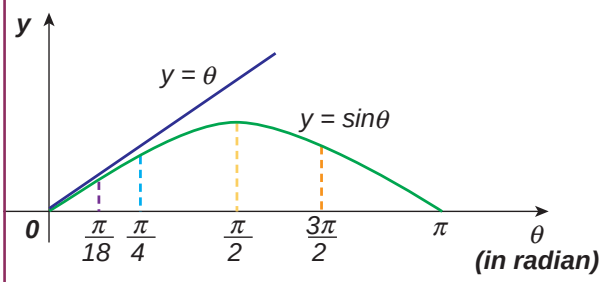
For a pendulum with small angle approximation (angular displacement is very small), the time period is independent of amplitude of the oscillation.

EXAMPLE 10.13

In simple pendulum experiment, we have used small angle approximation . Discuss the small angle approximation.

θ (in degrees)	θ(in radian)	sin θ
0	0	0
5	0.087	0.087
10	0.174	0.174
15	0.262	0.256
20	0.349	0.342
25	0.436	0.422
30	0.524	0.500
35	0.611	0.574
40	0.698	0.643
45	0.785	0.707

For θ in radian, $\sin\theta \approx \theta$ for very small angles



This means that "for θ as large as 10 degrees, $\sin\theta$ is nearly the same as θ when θ is expressed in radians". As θ increases in value $\sin\theta$ gradually becomes different from θ

Pendulum length due to effect of temperature

Suppose the suspended wire is affected due to change in temperature. The rise in temperature affects length by

$$l = l_o(1 + \alpha\,\Delta t)$$

where l_o is the original length of the wire and l is final length of the wire when the temperature is raised. Let Δt is the change in temperature and α is the co-efficient of linear expansion.

Then, $T = 2\pi\sqrt{\frac{l}{g}} = 2\pi\sqrt{\frac{l_0(1+\alpha\Delta t)}{g}}$:

$$= 2\pi\sqrt{\frac{l_0}{g}}\sqrt{(1+\alpha\Delta t)}$$

$$T = T_0\,(1+\alpha\Delta t)^{\frac{1}{2}} \approx T_0\,(1+\frac{1}{2}\alpha\Delta t)$$

$$\Rightarrow \quad \frac{T}{T_0} - 1 = \frac{T-T_0}{T_0} = \frac{\Delta T}{T_0} = \frac{1}{2}\alpha\Delta t$$

where ΔT is the change in time period due to the effect of temperature and T_0 is the time period of the simple pendulum with original length l_0.

EXAMPLE 10.14

If the length of the simple pendulum is increased by 44% from its original length, calculate the percentage increase in time period of the pendulum.

Solution

Since

$$T \propto \sqrt{l}$$

Therefore,

$$T = \text{constant}\ \sqrt{l}$$